

# Fully digital workflow for the fabrication of occlusal stabilization splints based on individual mandibular movement

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## ABSTRACT

**Objective:** This study was conducted to present a completely digital workflow for the fabrication of occlusal stabilization splints using CAD/CAM systems and a digital face bow based on optical sensor technology.

**Methods:** Digital scans of the maxillary and mandibular arches of 20 volunteers were obtained using an intraoral scanner. Jaw relation and mandibular movements were recorded with a digital face bow via optical sensors. The virtual increase of the vertical dimension of occlusion (VDO) was then performed, after which computer-aided design (CAD) of the occlusal stabilization splints was carried out. The corresponding splints were then manufactured using digitally controlled technology.

**Results:** A completely digital workflow for the manufacturing of occlusal stabilization splints was found to be clinically feasible. The corresponding data analysis revealed high congruence between virtual and physical occlusal contacts on the occlusal splint. Moreover, the appropriate guidance of the anterior teeth area was easily obtainable, and the time for adjusting the occlusion was less.

**Conclusions:** This study demonstrated that the fabrication of occlusal stabilization splints using a fully digital workflow is feasible. Compared to traditional impression-based manufacturing, several advantages of digital manufacturing include easy accessibility, time-efficient manufacturing, high-level accuracy in splint quality, and potential to manufacture duplicate splints.

**Clinical significance:** The proposed fully digital approach may help young dentists fabricating stable occlusal splints with beneficial curative effects. Meanwhile, it could also improve the production efficiency of stable occlusal splints, saving time for both doctors and patients while reducing labor costs.

## 1. Introduction

Temporomandibular disorders (TMDs) are among the top four clinical disorders that have the highest prevalence in stomatology. The incidence rate of TMD in the general population ranges from 28 to 88 percent, with the highest prevalence and confirmed diagnosis rate in the population aged 20–30 years, of whom most are women [1,2]. Non-invasive, conservative TMD treatment options that are commonly used include medical instruction, physical therapy, splint therapy, manipulation, biofeedback therapy, and drug and psychological therapy [3–5]. Since occlusal dysfunction has been considered to be dominant by most researchers among other pathogenic factors for TMD, occlusal splint therapy based on occlusal dysfunction has received widespread attention and acceptance among researchers [6–9].

Occlusal stabilization splints have been proven to be most useful

among other types of splints in the prevention and management of stomatognathic system dysfunctions [10–12]. These splints can significantly alleviate TMD symptoms due to their ability in helping the stomatognathic system attain neuromuscular balance and reduce abnormal muscle activities [13–15]. A cohort study conducted by Sletten, who studied 157 patients with symptoms of TMD, found that most patients' conditions were significantly improved when treated with an occlusal stabilization splint [16]. Alajbeg discovered that enzymes in saliva decreased when they were chosen as oxidative stress biomarkers in the oral cavity of organisms in order to quantitatively evaluate the effect of occlusal stabilization splints on pain relief in patients with chronic TMD, demonstrating the efficacy of occlusal stabilization splints [17].

A maxillary occlusal stabilization splint should be fabricated in order to ensure that there are balanced and simultaneous occlusal contacts of the mandibular teeth on the splint surface when closing the mandible, as

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well as canine guidance during all excursive jaw movements (lateral and protrusive). In addition, the occlusal plane of the occlusal splint, which should ideally be 2 mm higher than the central fossae of the second molars, must be uniformly flat and free of any crisscross contact between the tooth cusps and pits [18,19].

Traditionally, the occlusal splint is made of a denture base resin, which consists of a wax pattern fabrication as well as boxing, gluing, and heating of the occlusal splint, a process that is complicated and time-consuming [20,21]. Meanwhile, the traditional method in ascertaining jaw relation is to manually increase the vertical dimension of occlusion. Here, occlusal recording material may affect the mandibular position, and technical sensitivity is high [22]. Moreover, pore formation, a high residual monomer content, and shrinkage during pressing can impair the dimensional stability and accuracy of the occlusal splint [23]. The low-quality product produced using the traditional method means that it is vulnerable to significant deviations between the occlusal splint and original design, which is manifested by nonuniform occlusal contacts of the mandibular teeth on the splint in conjunction with inadequate or excessive protrusive and lateral guidance. Since it is difficult to fulfill the objective of the original design, more chair-side time is required for occlusal adjustment, thus impairing the efficacy of the splint.

Recently, revolutionary changes have taken place in regard to traditional design and manufacturing due to rapid advances in digital technology in prosthodontics [24–26]. Such changes have been widely applied in the design and fabrication of different kinds of oral prostheses, such as fixed prostheses, implant dentures, removable dentures, and complete dentures, thus making more treatment options available for dental practitioners [27,28]. Digital technology can streamline the traditional production process and address problems related to the plethora of errors and extended production times currently present in traditional impression-recording and lost-wax techniques, which could improve prostheses accuracy and reduce manufacturing times [29–31]. Moreover, patients feel that oral scanning is more comfortable than alginate impression [32].

As digital face bows have been widely used in clinical prosthodontics, numerous researchers have investigated digital face bows based on ultrasonic or optical sensor technology in the digital design of prostheses [33]. These studies have shown that the digital face bow can allow for the accurate acquisition of occlusal relationships and multi-angle dynamic reproduction of the mandibular movements of patients [34–37]. Accordingly, this study employs digital technology to design and fabricate an occlusal splint in order to ascertain its full potential in improving the manufacturing quality and dimensional stability of splints.

## 2. Materials and methods

### 2.1. Selection of subjects

A total of 20 volunteers (4 males and 16 females) were enrolled from the Department of Prosthodontics between August 2021 and December 2021. Informed consent forms were signed by the volunteers after an introduction to the clinical trial and the risks and benefits involved. This study has been reviewed by the Biomedical Ethics Committee of School of Stomatology (approval number: [2019]-R-76).

Inclusion criteria: (1) 18–40 years old; (2) complete dentition; (3) normal jaw opening; (4) intact tooth; (5) no occlusal disorder; stable jaw relation; and (6) no ongoing dental therapy, such as orthodontic or prosthodontic treatment.

Exclusion criteria: (1) pregnant/lactating women; (2) temporomandibular joint lesions found on clinical palpation or medical imaging examination; (3) jaw opening less than 3 fingers; (4) patients with occlusal dysfunctions; (5) patients with severe or moderate periodontitis; and (6) unable to undergo examination or treatment due to the presence of a psychological or mental disorder.

### 2.2. Experimental equipment and software

The main experimental equipment used in this study included an intraoral three-dimensional intraoral scanner (Trios3, 3Shape, Denmark; 3Shape TRIOS software 22.0), CAD/CAM milling machine (CORITEC 350, Germany), polymethyl methacrylate (PMMA) disk, and digital face bow (ModJaw, ModJaw SAS, France). The main software used was computer-aided design software (exocad DentalCAD 3.0, Exocad GmbH, Germany).

### 2.3. Digital occlusal splint generation

#### 2.3.1. Intraoral scanning data acquisition

A Trios3 intraoral three-dimensional scanner (3Shape, Denmark) was used to acquire digital impressions of the maxillary and mandibular arches, based on which digital models were generated. The model was then saved in the standard tessellation language (STL) format.

#### 2.3.2. Therapeutic jaw relation and mandibular movements acquisition

The data sets of the digital full dentition model were imported into the digital face bow (MODJAW, France). The volunteers were instructed to bite a cotton roll for 30 minutes to relax their muscles. The signal receiving device of the digital face bow was then installed on the volunteer's head and mandibular teeth (Fig. 1). The congruence between the positions of the teeth on the digital full dentition model and those on the real full dentition were then determined. The volunteers were instructed to perform small repeated jaw opening and closing movements, as well as mandibular protrusive and lateral movements. Their associated jaw relation and mandibular movements were recorded using the digital face bow. Simulation and multi-angle, sectional observations, as well as registration of the dentition and condyle points in static and dynamic occlusion, were then carried out with the help of ModJaw's virtual reality technology (Fig. 2). The range containing stable centric jaw relation was determined in the recorded data for small repeated jaw opening, after which the central fossae of the second mandibular molars were raised 2 mm and were selected as the target jaw relation (Fig. 3). The mandibular movements (protrusive and lateral) and the target jaw relation were matched with the help of the SET A NEW IMR module of the digital face bow. Following data acquisition, the datasets of the digital full dentition model were saved in STL format, while those of mandibular movements were saved in the extensible markup language (XML) format.

#### 2.3.3. Computer-aided design

The digital dentition model was imported with the target jaw position into EXOCAD. Next, the XML file of the mandibular movement track was personalized, the mandibular movement progress bar was dragged to control the digital mandibular model's movement, the virtual



Fig. 1. Signal receiving device of the digital face bow based on optical sensor technology.

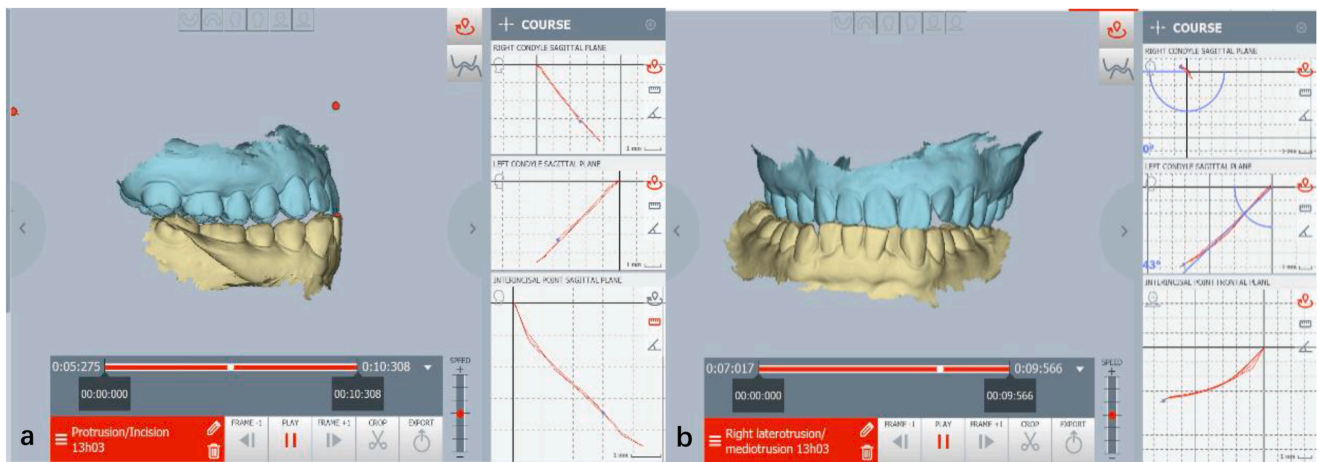


Fig. 2. Registration of dynamic mandibular movements a) protrusive movement; b) lateral movement.

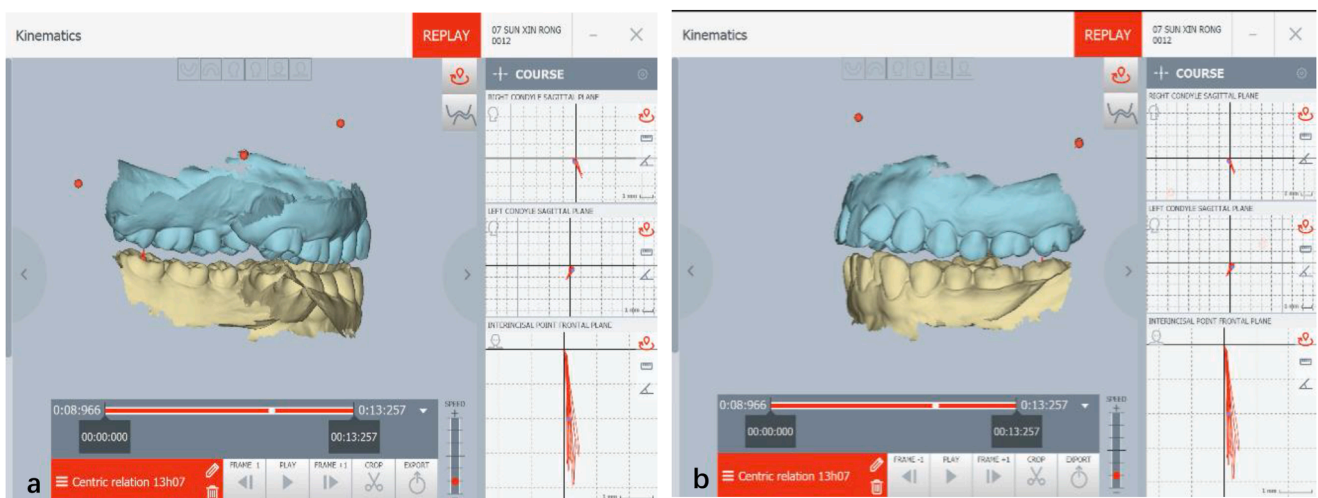


Fig. 3. New jaw relation after the central fossae of second molars were raised by 2 mm. a) lingual side; b) buccal side.

occlusion mark was observed, and the occlusal surface of the occlusal plate was designed. According to the design requirements of the stable occlusal plate, virtual occlusal adjustment was carried out, and the occlusal adjustment of dynamic movements, such as median occlusion, lateral movement and protrusive movement, was carried out in order to ensure that the functional tips of the whole dentition contacted with the occlusal plate at uniform points in the median relationship. Moreover, the canine area was sufficiently guided in the protrusive and lateral movement. A screenshot was then taken using EXOCAD to acquire the virtual bite mark on the bite plate (Fig. 4).

#### 2.3.4. Computer-aided manufacturing

Following CAD data acquisition, the occlusal stabilization splint was milled out of a PMMA disk using a denture milling unit (CORITEC 350) with a milling precision of 10  $\mu\text{m}$  [38].

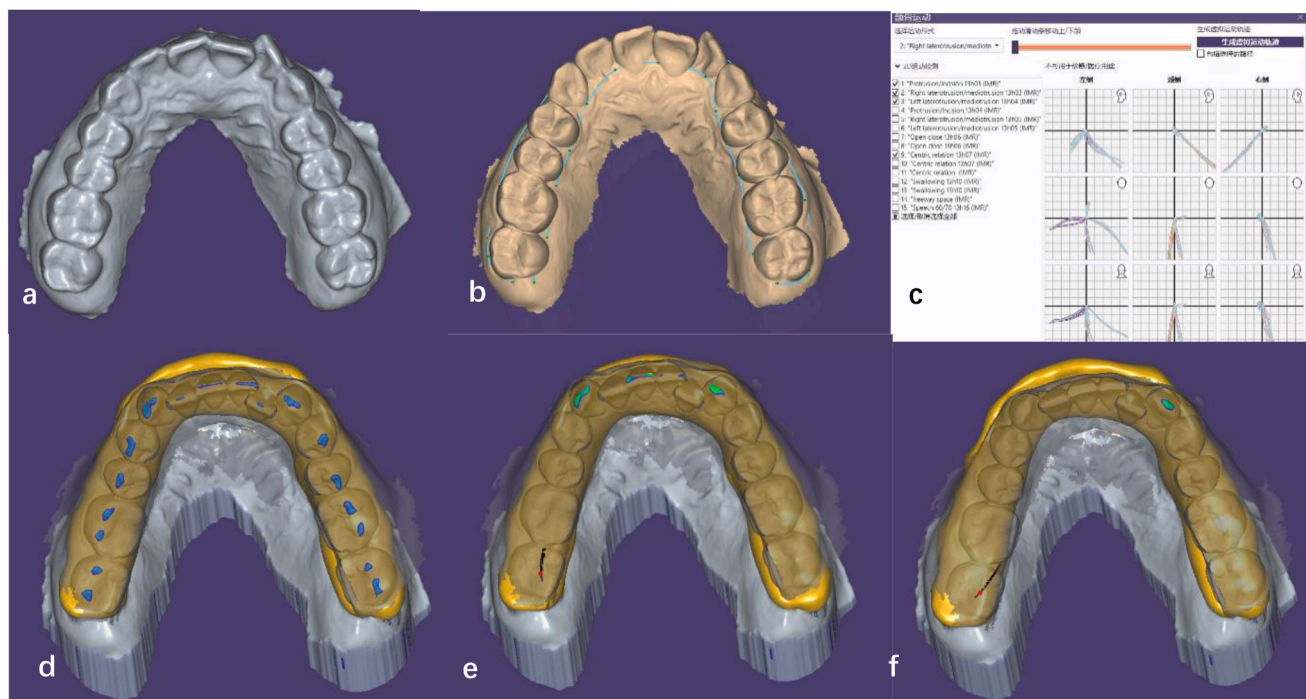
#### 2.4. Try-in of the occlusal splint

The volunteers were then instructed to stay in an upright seated position on the dental chair and keep their Frankfort plane parallel to the ground with their waists and backs slightly touching the back of the dental chair and eyes staring straight at a horizontal line on the wall two meters in front of them. The occlusal splint was inserted into the volunteers' maxillary dentition once their teeth surface was blown dry. After confirming that the occlusal splint was fully inserted into the

maxillary dentition, the volunteers were instructed to perform natural jaw opening and closing movements in order to ensure the mandibular dentition and occlusal splint were in uniform contact (Fig 5). Horseshoe-shaped, red articulating paper (Bausch, Germany) was subsequently used to determine the number of occlusal contacts between the mandibular dentition and occlusal splint. These occlusal contacts were then matched with virtual occlusal contacts to determine the congruence between them (Fig 6).

#### 2.5. Image overlapping and processing

Adobe Photoshop CC 2018 was used for graphics processing. The virtual occlusal mark image was opened as layer 1, while the real occlusal mark image was opened as layer 2, and the transparency was set as 85%. The two layers were then aligned and overlapped so that the edges of the occlusal plate shape in the two images overlapped onto the same position. Next, the lasso tool was used to select the virtual occlusal mark area on layer 1. In the real occlusal mark area on layer 2, the lasso tool was used to select the part that coincided with the virtual occlusal mark in layer 1. The selected area in the virtual occlusal plate was filled in blue, while the selected area in the actual occlusal plate was filled in red (Fig. 7). When the virtual and real occlusal contact areas of the same tooth were partially or completely overlapped, it was recorded to be overlapping. In doing so, the overlapping rate of the virtual occlusal mark and real occlusal mark can be calculated.



**Fig. 4.** Steps for designing digital occlusal stabilization splint. a) determination of the direction of insertion, and fill and smooth the undercut area; b) determination of the edge line; c) data of dynamic mandibular movements imported; d) occlusal adjustment; e) determination of protrusive movement guidance; f) determination of lateral movement guidance.



**Fig. 5.** The occlusal stabilization splint inserted into the volunteer's mouth.

### 3. Results

The fully digital workflow for manufacturing an occlusal stabilization splint was found to be clinically feasible, as shown in the workflow chart in Fig. 8.

SPSS 21.0 was used for data analysis and arrangement. The measurement data were described by the mean  $\pm$  standard deviation (mean  $\pm$ SD). Comparisons between groups were analyzed using the variance F test, and the difference was considered to be statistically significant when  $P < 0.05$ . Table 1 shows the overlapping rate of virtual occlusal contacts and actual occlusal contacts of different parts of the observation objects in the 20 cases. The overlap rate of the actual occlusal contacts of the entire dentition on the occlusal plate and the virtual designed occlusal contacts reached  $87.74 \pm 8.97\%$ . Meanwhile, the overlap rate of the occlusal contacts in the anterior area was found to be  $87.50 \pm 11.94\%$ , while that of the occlusal contacts in the premolar area was  $88.075 \pm 11.12\%$  and that of the occlusal contacts in the molar area was  $87.81 \pm 10.74\%$ . The variance F test demonstrated no significant difference in the overlap rate of the contacts in the anterior, premolar and molar areas ( $P > 0.05$ ).



**Fig. 6.** a) Virtual occlusal contacts; b) physical occlusal contacts; c) image overlay.



Fig. 7. Virtual occlusal mark (blue) and real occlusal mark (red) within the virtual occlusal mark area.

4. Discussion

The findings of this study demonstrated the feasibility of fabricating occlusal stabilization splints using a completely digital workflow. Although many researchers have introduced different methods for digital fabrication of occlusal splints, [39,40] they have certain limitations. First, existing methods used to manufacture occlusal splints are not

completely digital. For example, jaw relation acquisition is usually manually performed by doctors, which is a technically demanding task. Since there is a need for occlusal recording of silicone rubber or other media to record the occlusal relationship, patients will have some occlusal discomfort, which may lead to mandibular position deviation. Second, the parameters used in occlusal adjustment are the average values in the general population rather than personalized ones. Therefore, the present study built on the semi-digital manufacturing process of "scan-design-manufacturing" and presented a completely digital workflow to create occlusal splints by integrating two new processes: the digital registration of therapeutic jaw relation and virtual occlusal adjustment using personalized mandibular movements. This novel approach is less technical for doctors and can increase the accuracy of occlusal contacts.

Digital stomatology technology has become the main trend in prosthodontics. The widespread application of its corresponding software and hardware has offered new opportunities for digital technology in the

Table 1  
Comparison of the overlap rate of virtual and real contacts in different parts of the observed objects in 20 cases (%).

| Part             | Min   | Max    | mean±SD       |
|------------------|-------|--------|---------------|
| Entire dentition | 64.71 | 100.00 | 87.74 ± 8.97  |
| Anterior area    | 66.67 | 100.00 | 87.50 ± 11.94 |
| Premolar area    | 50.00 | 100.00 | 88.75 ± 11.12 |
| Molar area       | 57.14 | 100.00 | 87.81 ± 10.74 |

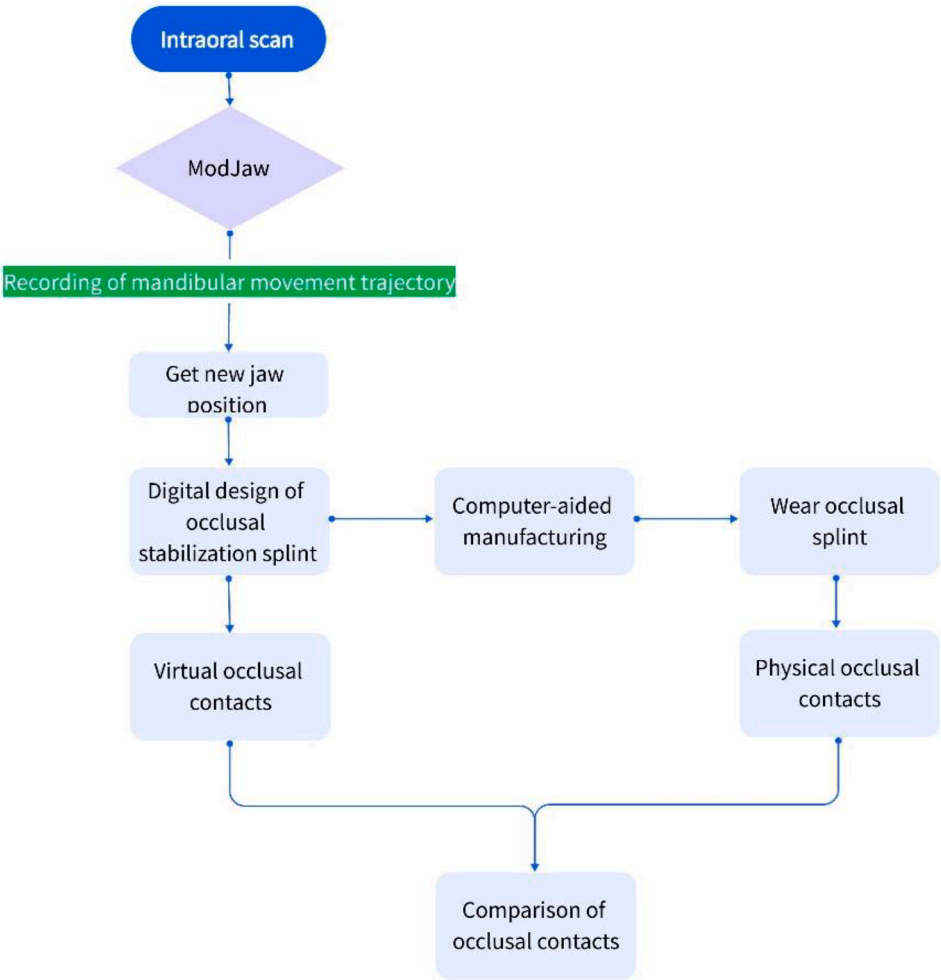


Fig. 8. The working flow chart of the whole digital manufacturing of the stable occlusal plate.

precise management of oral diseases, addressing some of the limitations of traditional treatment methods. In fact, digital technology has been used for the construction of a range of dental prosthesis, such as fixed prostheses, removable dentures and implantable dentures following considerable research and clinical application [41,42]. In all of these cases, digital technology has demonstrated desirable treatment outcomes and improved clinical efficiency in light of its modular and intelligent operation.

There are two types of CAM manufacturing: additive manufacturing (SLA) and subtractive manufacturing (SM), with SM having higher fabrication precision than that of SLA [43–45]. Therefore, subtractive manufacturing (numerical control milling using highly cross-linked PMMA disks) has been used in this study to increase the fabrication accuracy of occlusal splints by eliminating the shrinkage or deformation caused by the curing reaction during pressing.

Using digital techniques can also produce more aesthetically pleasing occlusal splints compared to using traditional methods. When a patient's vertical dimension of occlusion (VDO) is increased, the shape, width and color of the occlusal splint are almost identical to those of the patient's teeth [46], addressing the problem of poor medical order compliance caused by the unnatural appearance of the occlusal plate.

In terms of efficiency, intraoral scanning time is more time-consuming than taking traditional alginate impressions; the time required for the acquisition of mandibular movements and therapeutic jaw relation is comparable to that for obtaining therapeutic jaw relation using traditional methods [47]. It takes about 10 to 15 minutes from the start of intraoral scanning to the measurement of the patient's personalized mandibular trajectory, after which virtual elevation of the occlusal vertical distance is finally done to obtain the target jaw position. Nevertheless, traditional methods are more demanding for doctors; thus, the acquisition process may be prone to failure for inexperienced doctors, resulting in reduced efficiency in manufacturing occlusal splints.

Traditionally, the acquisition of mandibular movements was performed with a mechanical articulator, which simplified mandibular movements to linear ones, [48] resulting in reduced accuracy of acquisition. This limitation can be overcome by employing the digital face bow, as demonstrated by this study, which can acquire personalized mandibular movements through a fully integrated workflow. Specifically, ModJaw was used to record the data of mandibular posterior teeth knocking, mandibular protrusion and mandibular lateral movement, with ICP as the starting and endpoints. Following measurement, the mandibular snapping movement was repeatedly played back in the mandibular movement analysis system in order to find a suitable therapeutic jaw position. Essentially, the upper and lower dentition were unlocked from the cusp fossa, and the movement track of the condyle showed that the condyle point was still near the origin, whose data were associated with raising the vertical distance of the occlusion. Second, the trajectory of mandibular protrusion and lateral movement recorded by ModJaw was used for the virtual occlusal adjustment of the occlusal plate, including the design of anterior teeth guidance and posterior teeth separation during protrusion, as well as the design of canine guidance during the lateral movement.

This was shown to allow for significant improvement in virtual occlusal adjustment. Specifically, the early contact points of the posterior teeth in the protrusive movement and those on the non-working side in the lateral movement can be removed, thus reducing the possibilities of occlusal interference during try-in and decreasing the time needed to make occlusal adjustments. In fact, the time required for CAD/CAM splint fabrication is several times less than that of conventional impression-based methods [39].

In addition to reduced fabrication time, another major advantage of using a CAD/CAM system is the permanent availability of digitized data. Therefore, if an occlusal splint is lost or damaged, a duplicate can be easily made without the need for the patient to make an additional appointment for impression-taking. As labor costs are a major

component of current overall production costs, the reduction of labor time may greatly reduce costs, offsetting high acquisition costs in CAD/CAM systems.

All-digital workflows for occlusal stabilization splint fabrication have not yet been adopted on a large scale. One reason could be due to the high acquisition costs of CAD/CAM systems. In addition, many dental organizations are just starting to use such systems in their practice, and the price of the occlusal stabilization splints is relatively low, resulting in an unwillingness to risk losing money. However, the advent of more sophisticated scanning technologies will ultimately reduce the costs of CAD/CAM systems, which will accelerate large-scale adoption of all-digital workflows for splint fabrication in the future.

In this study, a new digital manufacturing process of occlusal plates was introduced; hence, no control group was established. In future studies, we hope to establish a control group to more finely verify the accuracy of the occlusal plate using the proposed method.

## 5. Conclusions

Occlusal stabilization splints with clinically acceptable quality can be manufactured using completely digital workflows. This approach makes it more convenient for doctors to acquire the therapeutic jaw relation and design a protrusive and lateral guidance, resulting in reduced time for occlusal adjustment. As higher fabrication efficiency and permanent availability of digitized data allow for significant savings in time for doctors, the proposed method could better serve the needs of patients in the future.

## CRedit authorship contribution statement

**Xinrong Sun:** Investigation, Methodology, Project administration, Software, Writing – original draft, Writing – review & editing. **Yue Feng:** Data curation, Formal analysis. **Yi Jiao:** Supervision, Validation, Visualization. **Weicai Liu:** Conceptualization, Funding acquisition, Resources.

## Declaration of Competing Interest

No potential conflict of Interest was reported by the authors.

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This study has passed the ethical review of the Ethics Committee of Stomatological Hospital of Tongji University, and has been approved by ethics.

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